

Linear Molecules in the Aliev Framework

Implemented equations, optical constant L , and frozen diagnostic conventions

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1. Scope

This note is a standalone record of the current linear-molecule implementation in `/Users/vincenzobarone/centrifugal/linear_dv_aliev_terms.py` and of the Gaussian bootstrap in `/Users/vincenzobarone/centrifugal/scripts/build_linear_aliev_payload_from_gaussian.py`.

The aim is limited and explicit:

- write all implemented linear Aliev equations in a clean form;
- include the quartic optical scalar L ;
- separate what is implemented algebraically from what is only frozen as a diagnostic convention.

This is not a claim that every remaining convention has been derived from first principles. It is the cleanest complete statement of the current linearmolecule branch.

2. Indices and objects

- Parallel modes are indexed by $n, n', n'', \dots$
- Perpendicular degenerate pairs are indexed by $t, t', \dots$
- B is the equilibrium rotational constant.
- D_J is the equilibrium quartic centrifugal distortion constant.
- ω_n and ω_t are harmonic frequencies of parallel and perpendicular modes.
- B_n^{xx} is the parallel rotational derivative entering $C_n = -BB_n^{xx}/\omega_n$.
- k'_{ijk} and k'_{ijkl} denote the cubic and quartic constants in the Aliev-side coordinate convention used by the implementation.

3. Primary definitions

The implemented primary objects are

$$C_n = -\frac{BB_n^{xx}}{\omega_n}, \quad (1)$$

$$X^{nt} = \frac{2B\zeta_{nt}\sqrt{\omega_n\omega_t}}{\omega_n^2 - \omega_t^2}, \quad (2)$$

$$X_{nt} = \frac{B\zeta_{nt}(\omega_n^2 + \omega_t^2)}{\sqrt{\omega_n\omega_t}(\omega_n^2 - \omega_t^2)}, \quad (3)$$

$$r_n = 4\omega_n C_n \left(\frac{D_J}{B} - \frac{B^2}{\omega_n^2} \right) + \frac{1}{2} \sum_{n', n''} k'_{nn'n''} C_{n'} C_{n''}, \quad (4)$$

$$r_{nn'} = \frac{3\omega_n\omega_{n'}C_nC_{n'}}{4B} + \frac{1}{2} \sum_{n''} k'_{nn'n''} C_{n''}, \quad (5)$$

$$r_{tt'} = \frac{1}{2} \sum_n k'_{tt'n} C_n. \quad (6)$$

4. Auxiliary families

The implemented F , U , and V families are

$$F^{nn'} = B^2 \sum_t \zeta_{nt} \zeta_{n't} \sqrt{\omega_n \omega_{n'}} \frac{\omega_n^2 + \omega_{n'}^2 - 2\omega_t^2}{(\omega_n^2 - \omega_t^2)(\omega_{n'}^2 - \omega_t^2)}, \quad (7)$$

$$F_{nn'} = r_{nn'} + B^2 \sum_t \zeta_{nt} \zeta_{n't} \frac{\omega_n^2 \omega_{n'}^2 - \omega_t^4}{\omega_t \sqrt{\omega_n \omega_{n'}} (\omega_n^2 - \omega_t^2)(\omega_{n'}^2 - \omega_t^2)}, \quad (8)$$

$$U_{nn'} = \frac{r_{nn'}}{4(\omega_n + \omega_{n'})} + \frac{B^2}{8} \sum_t \frac{\zeta_{nt} \zeta_{n't}}{\omega_t (\omega_n \omega_{n'})^{1/2}} \frac{(\omega_n + \omega_t)^2 (\omega_{n'} - \omega_t)^2 + (\omega_n - \omega_t)^2 (\omega_{n'} + \omega_t)^2}{(\omega_n^2 - \omega_t^2)(\omega_{n'}^2 - \omega_t^2)}, \quad (9)$$

$$V_{nn'} = \frac{r_{nn'}}{4(\omega_n - \omega_{n'})} + \frac{B^2}{8} \sum_t \frac{\zeta_{nt} \zeta_{n't}}{\omega_t (\omega_n \omega_{n'})^{1/2}} \frac{(\omega_n + \omega_t)^2 (\omega_{n'} + \omega_t)^2 (\omega_n + \omega_{n'} - 2\omega_t) - (\omega_n - \omega_t)^2 (\omega_{n'} - \omega_t)^2 (\omega_n + \omega_{n'})}{(\omega_n^2 - \omega_t^2)(\omega_{n'}^2 - \omega_t^2)(\omega_n - \omega_{n'})} \quad (10)$$

For the perpendicular branch the implementation also keeps the analogous families

$$U_{tt'} = \frac{r_{tt'}}{4(\omega_t + \omega_{t'})} + \frac{B^2}{8} \sum_n \frac{\zeta_{nt} \zeta_{n't'}}{\omega_n (\omega_t \omega_{t'})^{1/2}} \frac{(\omega_t + \omega_n)^2 (\omega_{t'} - \omega_n)^2 + (\omega_t - \omega_n)^2 (\omega_{t'} + \omega_n)^2}{(\omega_t^2 - \omega_n^2)(\omega_{t'}^2 - \omega_n^2)}, \quad (11)$$

$$V_{tt'} = \frac{r_{tt'}}{4(\omega_t - \omega_{t'})} + \frac{B^2}{8} \sum_n \frac{\zeta_{nt} \zeta_{n't'}}{\omega_n (\omega_t \omega_{t'})^{1/2}} \frac{(\omega_t + \omega_{t'})^2 (\omega_{t'} + \omega_n)^2 (\omega_t + \omega_{t'} - 2\omega_n) - (\omega_t - \omega_{t'})^2 (\omega_{t'} - \omega_n)^2 (\omega_t + \omega_{t'})}{(\omega_t^2 - \omega_n^2)(\omega_{t'}^2 - \omega_n^2)(\omega_t - \omega_{t'})} \quad (12)$$

5. Parallel coefficient β_n

The implemented parallel coefficient is

$$\begin{aligned} \beta_n &= 4\omega_n^2 C_n^2 \left(\frac{B}{\omega_n^2} - \frac{D_J}{B^2} \right) + \frac{1}{4} \sum_{n', n''} k_{\beta_n}^{(4)}(n; n', n'') C_{n'} C_{n''} - \sum_{n'} \frac{k'_{nnnn'}}{2\omega_{n'}} r_{n'} \\ &\quad - 4B \sum_{n', t} C_{n'} \zeta_{n't} \frac{(3\omega_n^2 + \omega_t^2) \omega_n^{3/2} C_n \zeta_{nt} + (\omega_n^2 + \omega_t^2) \omega_n^{3/2} C_n r_{nn'}}{\sqrt{\omega_n \omega_t} (\omega_n^2 - \omega_t^2)} \\ &\quad + 4 \sum_{n', t} \left(X_{nt} X_{n't} F^{nn'} + X^{nt} X^{n't} F_{nn'} \right) - 4 \sum_{n', t} \left(X_{nt} X^{n't} F_{n'n} + X^{nt} X_{n't} F^{n'n} \right) \\ &\quad - 16 \sum_{n'} [U_{nn'}(\omega_n + \omega_{n'}) + V_{nn'}(\omega_{n'} - \omega_n)]. \end{aligned} \quad (13)$$

Final working convention for the current linearmolecule branch. In the current working branch, the final parallel uv insertion is disabled:

$$(\beta_n)_{\text{frozen}} = (\beta_n)_{\text{full}} - \left[-16 \sum_{n'} (U_{nn'}(\omega_n + \omega_{n'}) + V_{nn'}(\omega_{n'} - \omega_n)) \right].$$

This is not presented as a literature-derived theorem. It is the final working convention of the present linear branch because, with the current Gaussian $\rightarrow$ Aliev mapping, the parallel uv block behaves as a dominant non-modal offset and drives otherwise sane parallel outputs to the wrong scale.

6. Perpendicular coefficient β_t

The implemented perpendicular coefficient is

$$\begin{aligned} \beta_t = & \frac{1}{4} \sum_{n,n'} k_{\beta_t}^{(4)}(t; n, n') C_n C_{n'} + \mathcal{P}_t - \sum_n \frac{k'_{ttn} r_n}{2\omega_n} \\ & - 4B \sum_{n,n'} C_n \zeta_{nt} \frac{2\omega_t^2 \omega_{n'}^{3/2} C_{n'} \zeta_{n't} + (3\omega_t^2 + \omega_{n'}^2) \omega_{n'}^{3/2} C_{n'} r_{nn'}}{\omega_t \sqrt{\omega_{n'}} (\omega_t^2 - \omega_{n'}^2)} \\ & + 4 \sum_{n,n'} \left(X_{nt} X_{n't} F^{nn'} + X^{nt} X^{n't} F_{nn'} \right) - 4 \sum_{n,t'} \left(X_{nt} X^{nt'} F_{tt'} + X^{nt} X_{nt'} F^{tt'} \right) \\ & - 16 \sum_{t'} [U_{tt'}(\omega_t + \omega_{t'}) + V_{tt'}(\omega_{t'} - \omega_t)]. \end{aligned} \quad (14)$$

Bending seed currently used. The currently frozen diagnostic branch uses

$$\mathcal{P}_t = \text{rotder_seed_gram},$$

namely the seed built from the geometry/mode-based rotational-derivative scaffold in `/Users/vincenzobarone/centrifugal/linear_aliev_rotder_zeta.py`.

Final working sign convention. After correcting the negative X/F block to use the perpendicular families $F_{tt'}$ and $F^{tt'}$ rather than reusing $F_{nn'}$ and $F^{nn'}$, the remaining bending discrepancy is still controlled by the common sign of the full perpendicular $xf/cross$ insertion. In the present linear branch this is encoded as

$$\text{beta_t_xf_cross_sign} = -1.$$

This is not yet a literature-proof derivation, but it is the final working convention reached after the remaining structural alternatives inside the implemented branch have been eliminated.

7. Optical scalar L

The implemented quartic optical scalar is

$$L = -\frac{8D_J^3}{B^2} + \frac{1}{24} \sum_{n,n',n'',n'''} k'_{nn'n''n'''} C_n C_{n'} C_{n''} C_{n'''} + 8BD_J \sum_n \frac{C_n^2}{\omega_n} - \sum_n \frac{r_n^2}{2\omega_n}. \quad (15)$$

This is exposed programmatically by

$$\text{build_explicit_aliev_L_model(inputs).value.}$$

The implementation also provides a bookkeeping partition of L into modewise contributions, but this partition remains provisional because the global $-8D_J^3/B^2$ term is not intrinsically mode-local.

8. Final state dependence

The state dependence implemented for linear molecules is

$$D_v = D_J - \sum_n \beta_n \left(v_n + \frac{1}{2} \right) - \sum_t \beta_t (v_t + 1). \quad (16)$$

9. Working linear branch

The current working branch for linear molecules is:

- bending seed:

`pair_seed_source = rotder_seed_gram;`

- perpendicular $xf/cross$ sign:

`beta_t_xf_cross_sign = -1;`

- parallel uv insertion:

$(\beta_n)_{\text{frozen}} \text{ uses } uv_{\parallel} = 0.$

With this branch the two current linear benchmarks in the repo behave as follows:

C₂H₂.

$$\begin{aligned} v_1 &\sim -1.530 \times 10^{-9}, & v_2 &\sim -1.199 \times 10^{-10}, & v_3 &\sim -9.539 \times 10^{-10}, \\ v_4 &\sim +5.970 \times 10^{-9}, & v_5 &\sim +4.728 \times 10^{-9}. \end{aligned}$$

HCN.

$$\beta_{\parallel} \sim 10^{-17} \text{ to } 10^{-15}, \quad \beta_{\perp} \sim -1.79 \times 10^{-15}.$$

The important point is not that these are yet a final literature match. The important point is that the catastrophic scale failures of the earlier branch are gone.

10. Honest status

The current linear branch should be read in the following way.

- The algebraic implementation of the linear families, of β_n , β_t , and of the optical scalar L is now explicit and internally consistent.
- The Gaussian bootstrap is no longer using the Gaussian Coriolis tensor as if it were already the physical Aliev ζ object.
- The present linear branch is numerically sane and suitable as a working benchmark branch.
- It is still not a fully derived, literature-validated final theory branch, because two conventions remain final working conventions rather than proved:
 - the sign of the perpendicular $xf/cross$ insertion;
 - the exclusion of the parallel uv insertion.

So the correct conclusion is:

The linear-molecule branch is now closed as a strong working implementation, including the optical scalar L , but not yet closed as a fully justified final derivation of every remaining convention.